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Shallow groundwater characterization for small-scale irrigation in the Northeastern Amhara, Ethiopia

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Abstract

This study assesses the aquifer characterization for small-scale irrigation in the semi-arid Northeastern Amhara Region, Ethiopia, focusing on the Kobo, Girana and Cheffa districts. The study involved the integrated techniques of geophysical survey using Vertical Electrical Sounding employing Schlumberger array, field check of existing hand-dug wells, hydrological studies and soil properties analysis. The results indicate substantial shallow groundwater potential, *at Kobo, Girana, and Cheffa, where* farmers have successfully excavated hand-dug wells. The borehole data indicates at Kobo, Girana, and Cheffa exhibit static water levels at depths of 8 m, 5 m, and 10 m, with corresponding total drilled depths of 40 m, 13 m, and 32 m, respectively. The identified potential land was 862 ha of land suitable for shallow groundwater coverage, comprising 448 ha in Kobo, 200 ha in Girana, and 214 ha in Cheffa. However, there are significant gaps between potential and actual use with only 220 ha and 150 ha irrigated in Kobo and Cheffa respectively. The pumping discharge data from 25 boreholes in Cheffa indicated a uniform yield of 16.6 L/sec with a total annual capacity of about 130,994 m³. The cumulative discharge rates from the Kobo and Girana well field indicated the opportunity for substantial agricultural development. The optimized development of shallow groundwater could provide for up to three growing seasons annually, contributing significantly to food security and agricultural productivity. These results provide critical scientific evidence for policymakers, planners, and local communities to optimize groundwater utilization while ensuring long-term sustainability in this drought-prone region.

Keywords Aquifer characterization, Irrigation potential land, Kobo, *Girana*, Cheffa, Northeastern Amhara, *Ethiopia*

1 Introduction

Groundwater is a critical resource for agricultural production, particularly in semi-arid regions where surface water is scarce and unreliable [1]. Globally, groundwater supplies approximately 40–43% of irrigation water, supporting the agricultural production of billions of people living in water-scarce environments [2, 3]. In Ethiopia, agriculture accounts for approximately 35% to GDP and 68.2% to employment and 90% of the



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country's export earnings, making water availability a key factor in the nation's economic development [4]. In the semi-arid regions of northeastern Amhara, specifically in the Kobo, Girana, and Cheffa districts, agriculture is heavily reliant on groundwater resources to supplement rain-fed farming systems, which are threatened by increasing variability in precipitation due to climate change.

The reliance on shallow groundwater systems for irrigation has become increasingly important in North and South Wollo, Oromia administrative zones. Shallow groundwater sources are typically found within 35 m of the subsurface and are accessible through shallow wells and hand-dug systems, making them more feasible and cost-effective for smallholder farmers [5]. However, shallow groundwater resources in Ethiopia are poorly characterized and inadequately managed. While deep aquifers have been the focus of many studies and development projects, shallow groundwater resources remain under-explored, which can potentially lead to unsustainable water use and environmental degradation [6]. A systematic assessment and characterization of shallow groundwater potential in the Kobo, Girana, and Cheffa districts is critical for optimizing irrigation potential, ensuring sustainable use, and supporting agricultural productivity in the region. The aim of this study is to delineate accessible shallow groundwater aquifers for manual excavation through geophysical mapping, specifically electrical resistivity; and to promote sustainable, modern irrigation technologies that optimize water use during dry periods. By focusing on user-friendly systems that provide reliable, on-demand water access, this research will make practical recommendations for local communities and strategic guidance for policymakers on enhancing water resource management. In Ethiopia's Northeastern Amhara region, specifically in the districts of Kobo, Girana, and Cheffa, agriculture is the predominant livelihood, and irrigation is a vital component for food production. Though, the recurring droughts and erratic rainfall patterns in this region necessitate the exploration of shallow groundwater resources to support irrigation.

This study aims to assess and characterize shallow groundwater potential in the Kobo, Girana, and Cheffa districts. The study focuses on identifying areas with significant groundwater potential and evaluating the suitability of these resources for irrigation, ensuring sustainable water management. The results of this study inform water resource managers, farmers, and policymakers on the best strategies for harnessing aquifer characterization for sustainable irrigation, ultimately contributing to the region's agricultural development and resilience to climate change.

2 Study area description

The study focuses on the North Wollo, South Wollo and Oromia administrative zones, especially the Kobo, Sirinka, Girana, Jari and Cheffa districts, in the northeastern part of the Amhara region, Ethiopia, geographically situated between latitudes 11°30' and 12°50' N and longitudes 39°40' and 40°45' E. The area spans across a wide range of altitudes, from around 1400 m above sea level in the lowland areas of Kobo to more than 2500 m in the highlands of the Cheffa district (Fig. 1). This elevation gradient influences not only the climate but also the surface and subsurface hydrological processes that govern groundwater recharge and availability. This region, positioned within the upper reaches of the Awash River Basin, is characterized by diverse topographical, climatic, and geological features. These variations play a crucial role in determining groundwater

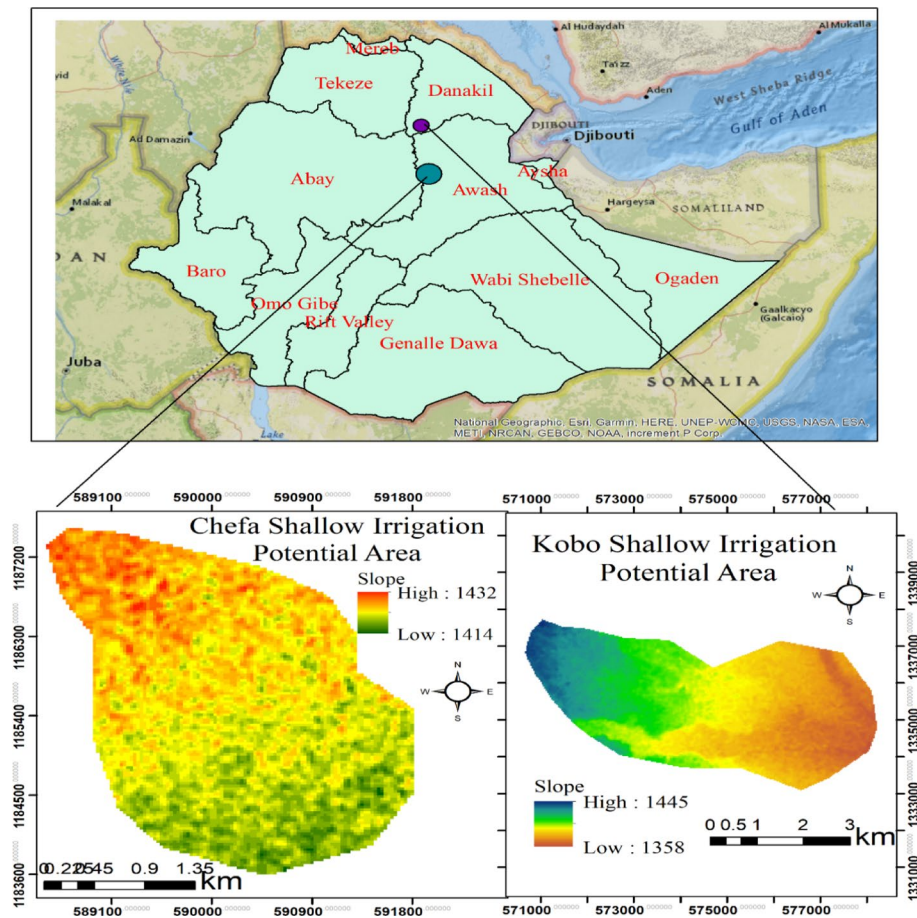


Fig. 1 Map of the study areas

potential and have a flat and highly fertile land for agriculture. The topography, drainage patterns, and soil properties are the primary factors influencing groundwater availability. The northeast Amhara region areas are semi-arid and uneven distribution of rainfall and high susceptibility to drought.

3 Materials and methods

3.1 Subsurface hydrogeological study

Geophysical investigation of the subsurface structure was studied by using ABEM Terrameter instrument. This system is essential for acquiring critical subsurface structural information by deploying electrodes into the ground and measuring the electrical properties of the underlying geological formations [7, 8]. The instrument works on the basic principles of electrical resistivity tomography and induced polarization using to inject electrical current through the ground while measuring any resulting potential differences across electrodes. The electrical resistivity survey employs electrical sounding utilizing the Schlumberger setup with an ABEM Terrameter. The Schlumberger array used, with maximum current electrode separation of $(AB/2)1000$ m, electrodes are normally arranged along a straight line, with the potential electrode placed between the current electrodes. This configuration is mostly used subsurface information considering the approximate depth of investigation is about one-third to one-half of the current electrode spacing depending on subsurface conditions [9, 10]. Using the VES approach,

we were able to determine the underlying soil thickness, depth, and texture as well as the overall hydrogeological structure. In areas underlain by unconsolidated sediments, manually excavated wells can reach depths of approximately 30–35 m where bedrock is absent [11]. This is possible because the unconsolidated sediment profile permits manual excavation. The subsurface data were collected using vertical electrical survey (VES) in Kobo (Aborie), Sirinka (Gerado), Girana, Hike (Jari), and Cheffa (Kachure).

One of the field measurements techniques were digital water level meters, used to measure groundwater levels from existing boreholes and observation wells. Measurements were taken at regular intervals at irrigation season to monitor fluctuations over time.

3.2 Soil properties

Soil texture, hydraulic conductivity, porosity, and moisture retention characteristics were determined through field surveys and laboratory analyses. These properties play a critical role in assessing the infiltration rates and groundwater recharge capacity of the soils in the study area. Soil samples were analyzed for texture, moisture content, permeability, and hydraulic conductivity using standard laboratory procedures. The results provided essential data for assessing soil-water relationships and infiltration rates.

3.3 Meteorological data

This study used a long-term meteorological dataset to examine the climate patterns of the Kobo, Habru, and Cheffa Boundary region. Daily and monthly records of precipitation and temperature, spanning 35 years (1987–2021), were acquired from nine local stations of the Ethiopian National Meteorological Agency (ENMA). These stations were Woldia, Sirinka, Mersa, Hara, Dessie, Kombolcha, Cheffa, Majetie, and Kemesie which provided the fundamental information needed to describe the distribution of rainfall and temperature regimes in the region.

3.4 Irrigation potential

The Kobo, girana, and Cheffa districts have shallow groundwater resources that could be used to expand irrigated agriculture. To effectively harness this potential, a rigorous assessment and characterization of groundwater availability are essential. This work will generate a detailed agricultural suitability map focused on areas where aquifer characterization can be feasibly and sustainably developed for irrigation. The analysis will specifically highlight zones with high natural recharge and reliable groundwater availability, thereby delivering a scientific and spatially explicit basis for informed irrigation planning and sustainable water resource management.

3.5 Characterization of the study area

3.5.1 Climatic characteristics of the study area

The climate in the study area varies from semi-arid in the lowlands to sub-humid in the highland regions. Rainfall is highly seasonal, with the majority of precipitation occurring during the Kiremt season (June to September). Annual rainfall ranges from 700 mm in the drier, lower-lying areas of Kobo Girana to over 1200 mm in the higher altitudes of Cheffa. The variability in rainfall poses significant challenges for rain-fed agriculture, making the role of groundwater crucial for irrigation, especially during the dry

seasons. Temperature patterns also vary with altitude. In the lower areas, such as Kobo and Girana, mean temperatures range between 20 and 35 °C, while in the higher regions of Cheffa, temperatures are cooler, ranging from 10 to 25 °C. The high temperatures in the lowlands lead to higher evapotranspiration rates, increasing the need for effective groundwater management to supplement surface water resources for agricultural activities.

3.5.2 Rainfall distribution of the study area

The rainfall distribution of the three-study area (Habru, kobo and Cheffa Boundary) shows considerable spatial and temporal variability, influenced by its diverse topography and climatic zones. The three of the Boundary receives an average monthly rainfall ranging from minimum 0 mm to maximum 450 mm, with significant differences observed between the dry and wet seasons. During the dry season, typically from November to February, rainfall is minimal or completely absent, resulting in values close to 0 mm (Fig. 2).

3.5.3 Temperature

The temperature characteristics of the Habru, Dawa Chaffa, and Kobo boundary exhibit a significant variation due to the wide range of elevation, above sea level. In the lower elevations, particularly in the Kola climatic zones, temperatures are generally high, often exceeding 25 °C, and the climate is hot and dry for much of the year. As the elevation increases towards the midlands, which fall within the Wena Dega zone, the temperatures become moderate, typically ranging between 18 and 25 °C, creating favorable conditions for diverse agricultural activities. In the highland areas, corresponding to the Dega zone,

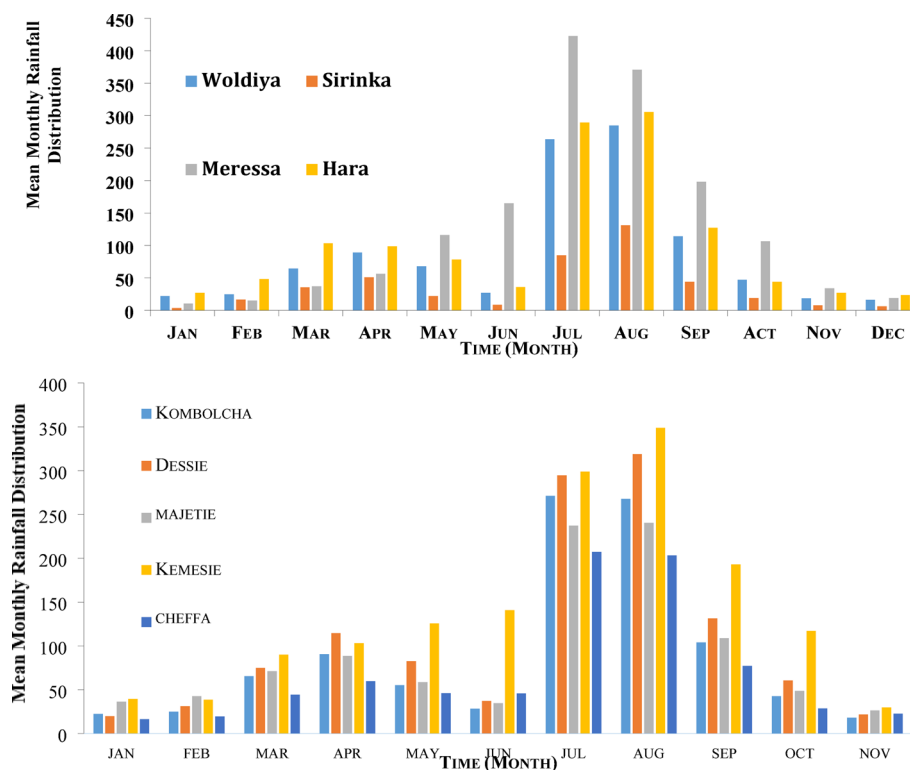


Fig. 2 Mean monthly rainfall distribution

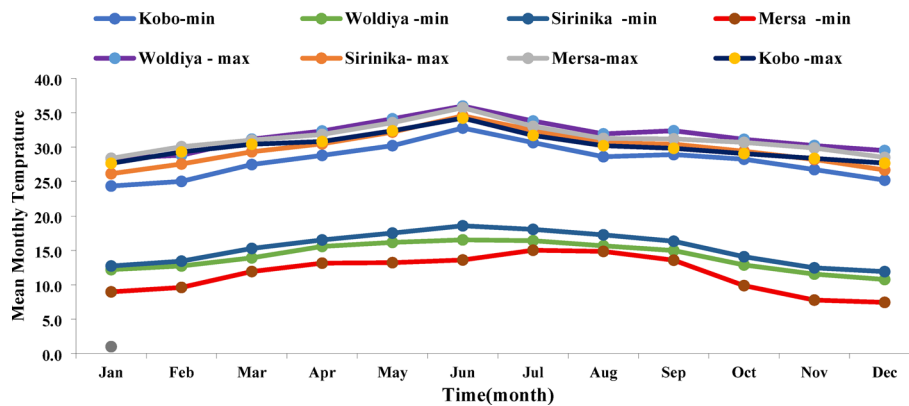


Fig. 3 Mean monthly maximum and minimum temperature distribution in the kobo boundary

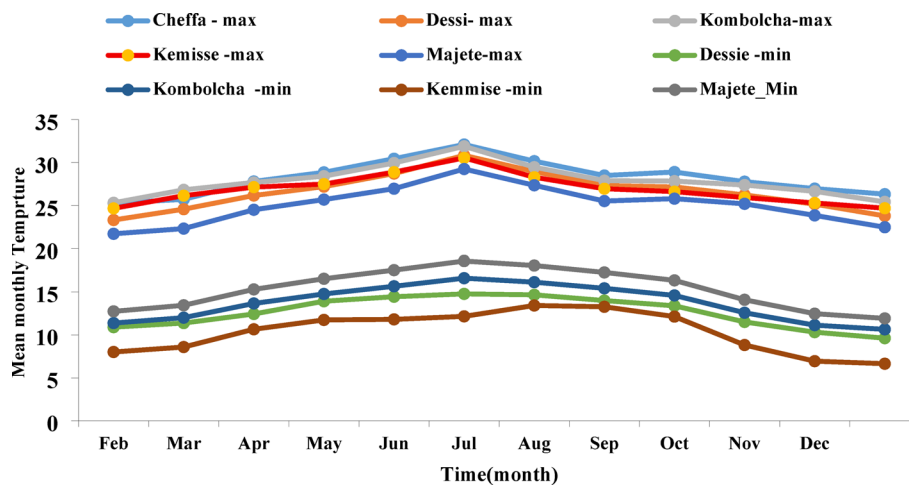


Fig. 4 Mean monthly maximum and minimum temperature distribution in the Dawa Cheffa

the climate is much cooler, with mean annual temperatures often falling below 15 °C. These highland regions experience shorter growing seasons and may occasionally face frost conditions. Overall, the three study areas demonstrate a distinct temperature gradient from the warmer lowlands in the east to the cooler highlands in the west, which significantly influences land use, vegetation cover, and water resource dynamics (Figs. 3 and 4).

3.5.4 Topography

The topography of the study area is characterized by a combination of flat plains, rolling hills, and mountainous terrains. The lowland plains of Kobo Girana, known for their relatively gentle slopes, provide favorable conditions for shallow groundwater accumulation. On the other hand, the hilly and mountainous regions of Mersa and Cheffa are defined by steeper slopes, which result in higher runoff and reduced groundwater infiltration. However, these regions serve as important recharge zones for groundwater due to the presence of fractured rocks and alluvial deposits that enhance infiltration. Several seasonal rivers and streams, including the Hormate, Borkena and Cheleka Rivers, traverse the study area, contributing to the surface water hydrology and potential groundwater recharge. These water bodies are vital for both agriculture and livestock, but they

often dry up during the long dry seasons, further emphasizing the importance of shallow groundwater as a reliable source of water for irrigation.

The topography of the Dawa Cheffa, Habru and Kobe Boundary's, as depicted in the elevation map, varies significantly, ranging from 963 to 3,142 m, 958 to 3389 m and 1162 to 2838 m above sea level for kobo, Habru and Dawa cheffa boundary respectively.

3.5.5 Geomorphology

The geological formations in the studied area vary greatly and have a considerable impact on the distribution and availability of groundwater resources. Volcanic rocks, alluvial deposits, and sedimentary strata comprise the region's foundation. Quaternary alluvial deposits dominate in the lower-lying regions of Kobo and Girana areas creating ideal conditions for shallow groundwater deposition [12]. This region is tectonically active, with normal faults and marginal grabens that shape drainage and groundwater storage. The geomorphology consists of Quaternary alluvium interfingering with fractured volcanic rocks, making the valley an important groundwater storage area. This pattern creates a sharp seasonality in water availability and agricultural production.

3.5.6 Groundwater recharge

The geology and ground water potential assessment at Girana and Kobo valley have been determined by the Commission for Sustainable Agriculture and Environmental Rehabilitation in Amhara (Co-SAERAR) in 1996 through 1999 and review and appraisal of Hydrogeological studies by Geo-Engineering Service in 2002/3. The study made preliminary estimation of annual recharge capacity of the Kobo and Gerana sub basins to be 119 and 27 MCM, respectively [13]. In 2015 hydrogeological study of the Kobo Valley, irrigation currently uses about 30,600 cubic meters of groundwater per day and the aquifer receives an average of 122.9 million cubic meters of natural recharge each year [12] and [14]. A comparison of these numbers shows that the rate of recharge is much higher than the rate of abstraction at the moment, indicating a sustainable water balance for current irrigation techniques. The river network with the headwaters of the Borkena River in the Kutaber highlands, function as *gaining* streams, predominantly fed to groundwater recharge and also the rivers near Harbu and Dessie are classified as *losing* streams, contributing to aquifer recharge. Based on the hydrological assessment of the Cheffa Valley indicates an estimated groundwater recharge rate of 370 mm per year [15].

4 Result and discussion

4.1 Geological characteristics

A geophysical evaluation using electrical resistivity method for groundwater investigation was carried out which involved the utilization of Vertical Electrical Sounding (VES) technique with Schlumberger array system [16]. The basic principle involves injecting current into the ground and measuring the resulting voltage to calculate resistivity [17]. The sub-surface geologic materials have different nominal resistivity, different materials have different resistivity, so by measuring this, we can infer subsurface structures [18]. Vertical Electrical Sounding (VES) was used to assess subsurface hydrogeology at five study sites: Kobo Abourie Kebele, Habru Geredo and Girana Kebele, Hyke Jari Agricultural Trial Site, and Dawa Cheffa Gobeya Kebele. The method effectively defined critical

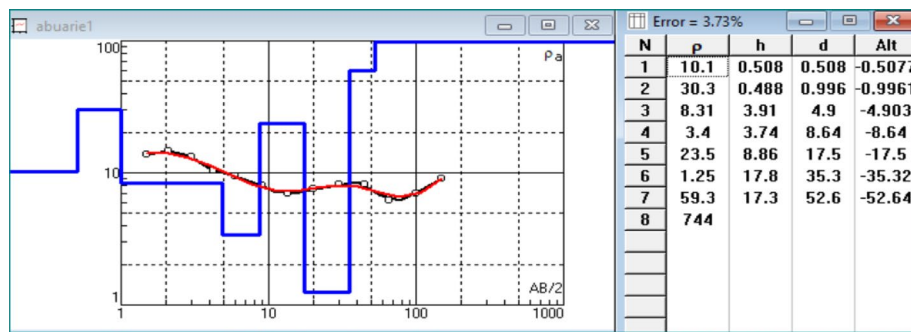


Fig. 5 Vertical electrical resistivity variation curve at kobo, abuarie kebele

Table 1 Description of resistivity, thickness, depth and its interpretation

N	Resistivity (Ω -m)	Thickness (m)	Depth (m)	Geological inference
1	10.1	0.508	0.508	Clayey topsoil
2	30.3	0.488	0.996	Sandy layer
3	8.31	3.91	4.9	Saturated zone (groundwater)
4	3.4	3.74	8.64	High-conductivity layer
5	23.5	8.86	17.5	Coarse sand/gravel
6	1.25	17.8	35.3	Highly conductive
7	59.3	17.3	52.6	Consolidated sandstone
8	744	–	–	Bedrock (granite)

factors such as soil thickness, depth of layers, soil texture, and overall hydrogeological structure.

4.1.1 Kobo

In kobo abourie kebele, the results of the VES survey show a progressive sequence that progresses from unconsolidated surficial deposits to a consolidated bedrock unit. A clayey topsoil layer that is 0.508 m thick at the beginning of the profile is followed by a sandy layer that is 0.488 m thick. The first two layers extend to a cumulative depth of 0.996 m (Fig. 5). Below this; the water table horizon is identified at a depth of about 4.9 m by a saturated zone that is 3.91 m thick. A noticeable high-conductivity layer that is 3.74 m thick and extends to 8.64 m further characterizes this aquifer (Table 1).

A notable 8.86-meter stratum of coarse sand and gravel (17.5-meter total depth) that is considered as a main aquifer unit follows the sequence's transition. A highly conductive layer extends from 17.5 to 35.3 m depth, indicating that fruitful hydrogeological conditions may continue down to 35.3 m. At the end a thick, 17.3-meter sequence of consolidated sandstone, forming the basal bedrock unit from 35.3 to 52.6 m depth. These results suggest that manually excavated wells may successfully access groundwater between approximately 8 and 28 m depth., the depth greater than 35 m will be effective for intended purpose.

4.1.2 Girana

The dataset represents a Vertical Electrical Sounding (VES) survey using the Schlumberger array to investigate subsurface geology. These layers resistivity to infer geological materials, resistivity values correlate with material types, low resistivity might indicate

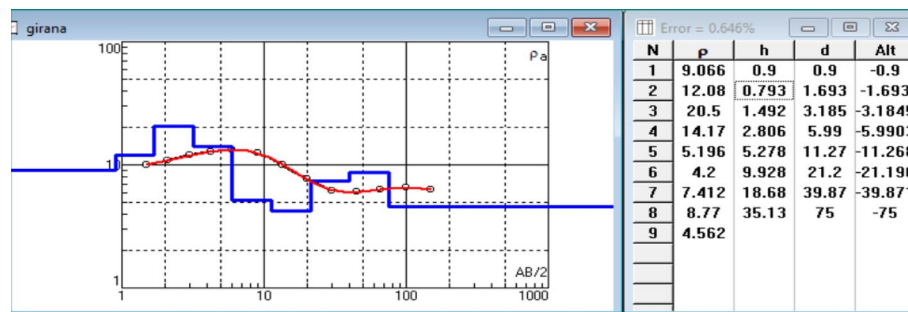


Fig. 6 Vertical electrical resistivity variation curve at Girana

Table 2 Description of resistivity, thickness, depth and its interpretation at Girana

N	Resistivity ($\Omega\cdot\text{m}$)	Thickness (m)	Depth (m)	Geological Interpretation
1	9.066	0.9	0.9	Silty clay/topsoil
2	12.08	0.793	1.693	Sandy silt
3	20.5	1.492	3.185	Dry sand/gravel
4	14.17	2.806	5.99	Moist sand-silt mix
5	5.196	5.278	11.27	Clayey layer
6	4.2	9.928	21.2	High-conductivity silt clay layer
7	7.412	18.68	39.87	Gravel and coarse sand
8	8.77	35.13	75	Coarse sand or weathered rock
9	4.562	-	> 75	Deep aquifer or clay-rich layer

clay that means unconfined aquifers, while high resistivity suggests sandstone or bedrock [19]. The (Fig. 6) shows resistivity values in each thickness with geological inferences.

Nine different lithological layers were found in the Grana Valley subsurface, according to resistivity profiling using a Schlumberger array [17, 20] Layer 1 ($9.066 \Omega\cdot\text{m}$) is speculated to contain damp topsoil or silty clay. Sandy silt and more resistive units (Layers 2, 3 and 4: $12.08 \Omega\cdot\text{m}$, $20.5 \Omega\cdot\text{m}$, $14.17 \Omega\cdot\text{m}$) which indicates of dry sand/gravel and mixed sand-silt were found beneath the surface. Layers 5 ($5.196 \Omega\cdot\text{m}$) and 6 ($4.2 \Omega\cdot\text{m}$) define a significant conductive zone, which corresponds to saturated clay. After that, resistivity rises through a transitional zone of coarser sediments (Layer 7 and 8, $7.412 \Omega\cdot\text{m}$, $8.77 \Omega\cdot\text{m}$) were gravel and coarse sand, (Layer 9, $4.562 \Omega\cdot\text{m}$) deeper conductive layer than other layers. The table below provides extensive geological information regarding the Grana Valley (Table 2).

These alluvial aquifers are typically unconfined [21], making them accessible for shallow wells and boreholes, which are often used for irrigation. In general, Below the surface there are silty clay or moist topsoil, sandy silt or gravel, and a mixture of sand and silt strata at a depth of 3.18 m. There is a high conductive clay layer and damp sand beneath 3.18 to 21.2 m, and gravel and coarse sand above 21.2 to 39 m.

4.1.3 Gerado

According to Schlumberger array system [17, 20]. The first layer the resistivity is $39.91 \Omega\cdot\text{m}$. That's relatively high, which indicate a more resistive layer dry sand or gravel (Fig. 7). The second layer has a much lower resistivity of $1.402 \Omega\cdot\text{m}$ which is low resistivity clay or saturated sediments. The third layer increases to $13.04 \Omega\cdot\text{m}$, a layer of less conductive material, saturated sandstone. The fourth layer resistivity drops to $5.649 \Omega\cdot\text{m}$ the thickness here is 6.694 m , leading to a cumulative depth of 8.41 m .

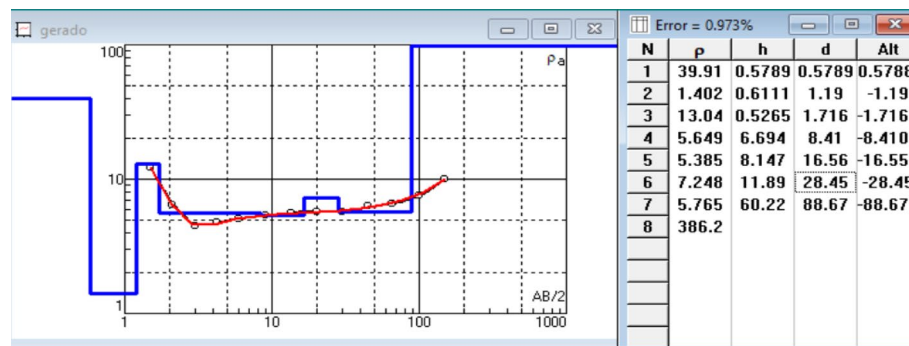


Fig. 7 Vertical electrical resistivity variation curve at Gerado (sirinka)

Table 3 Description of resistivity, thickness, depth and its interpretation Gerado (Sirinka)

Layer (N)	Resistivity [Ω -m]	Thickness [m]	Depth Range [m]	Material/Description
1	39.91	0.5789	0–0.58	Dry topsoil, clay and sand
2	1.402	0.6111	0.58–1.19	Clay and silt soil
3	13.04	0.5265	1.19–1.72	Sand saturated sediments
4	5.649	6.694	1.72–8.41	Clay with high water content
5	5.385	8.147	8.41–16.56	Homogeneous clay persistence
6	7.248	11.89	16.56–28.45	Coarser sediments sandy
7	5.765	60.22	28.45–88.67	Deep conductive aquifer clay
8	386.2	N/A	> 88.67	Bedrock basaltic formation

Lower resistivity indicate a more conductive layer, clay. But resistivity is depending on water content and composition. The fifth layer resistivity slightly decreases to 5.385 Ohm-m, thickness 8.147 m, depth 16.56 m, which is same material at previous layer but thicker. Layer 6 resistivity increases to 7.248 Ohm-m, thickness 11.89 m, depth 28.45 m. It is high resistivity which is a change in material, like more consolidated sediments or fractured bedrock with some moisture. Layer 7 resistivity drops again to 5.765 Ohm-m, but the thickness increases to 60.22 m, leading to a depth of 88.67 m, which means a significant increase in thickness and slight drop in resistivity might indicate bedrock like granite or limestone.

This could represent a typical sequence of topsoil, clay, sand ending with bedrock. The deposit thickness in that specific locality is not expected to be greater than 20 m because highly weathered basalt is exposed difficult to penetrate by hand-dug. As a result, layer 6, 7 and 8 will be the responses of basaltic formation (Table 3).

So manual shallow well drilling will be effective till depth less than 20 m, but aquifer till this depth will not be enough for irrigation purpose because it is not unconfined aquifer it might be perched aquifer.

4.1.4 Jari

According to Schlumberger array system [17, 20], the first layer has a resistivity of 14.8 Ω m, its thickness is 0.631 m and top soil formation indicates silt and clay soil (Fig. 8). The second layer has a much higher resistivity of 854 Ω m, with thickness 1.07 m, and depth 1.7 m. This is highly resistive, indicating low moisture content dry sand, gravel and compact sediment. The third layer indicates 197 Ω m, and its thickness is 0.972 m at a depth of 2.68 m. Reduced resistivity due to groundwater presence saturated sand/weathered rock. The resistivity decreases to 2.69 Ω m in layer four, then increases thicknesses

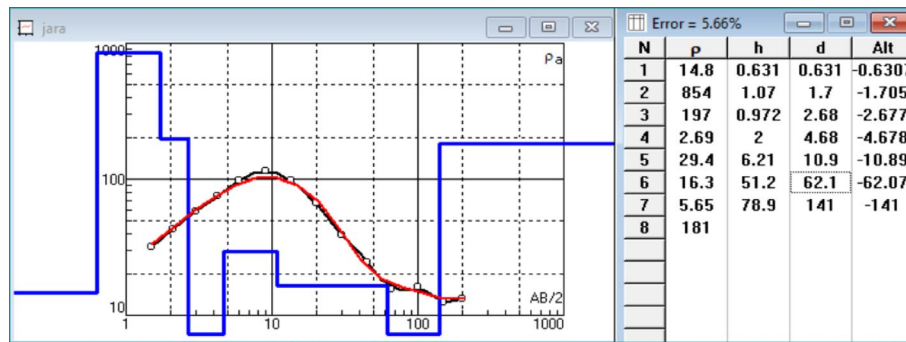


Fig. 8 Vertical electrical resistivity variation curve at jari

Table 4 Description of resistivity, thickness, depth and its interpretation at Jari

Layer	Depth Range (m)	Resistivity (Ω m)	Thickness (m)	Geological Information
1	0–0.63	14.8	0.631	Topsoil/Silty Clay
2	0.63–1.7	854	1.07	Dry Sand/Gravel
3	1.7–2.68	197	0.972	Sand/Weathered Rock
4	2.68–4.68	2.69	2	Clay
5	4.68–10.9	29.4	6.21	Sandstone/Unsaturated Gravel
6	10.9–62.1	16.3	51.2	Silt-Clay
7	62.1–141	5.65	78.9	Clay
8	> 141	181	Undefined	Bedrock/Compact Layer

and depths increase with each layer. The fifth layer depth ranges from 4.68 to 10.9 m it is highly resistivity 29.4 Ω m, coarse-grained sediments. The sixth layer depth ranges from 10.9 to 62.1 m, decreasing resistivity to 16.3 Ω m, it is a mix silt and clay formation. Layer seven resistivity values are 5.65 Ω m, very low, within 78.9 m thickness, 141 m depth. This could be a deep conductive clay layer. The substratum will be the signature of highly to moderately fractured basalt.

Even the groundwater table is expected to occur at relatively shallow depth but manual shallow well drilling will not be effective due to the presence of coble and boulder size deposit. Here is a summarized description of the geological layers based on the Vertical Electrical Sounding (VES) data at jari (Table 4).

4.1.5 Chaffa

The surface layer known as topsoil is thin and has a moderate resistance. It could be made of sand and loose dry earth. The thick clay layer may reduce vertical infiltration while enhancing groundwater storage in the underlying sandy deposits. An extremely thick, highly conductive (low resistance) layer is found in layer four. Because of its low resistivity, this is a certain sign of a clay-rich horizon that is saturated with water (Fig. 9). The resistivity in layer five suggests that the material may have changed to a sandy layer. This transition points to a change in the geological makeup, which could result in distinct hydrological traits. Assessing the area's soil stability and groundwater flow requires an understanding of these strata.

Additionally, recognizing these geological variations can aid in predicting how contaminants might move through the subsurface environment. The moderate resistivity suggests it contains more water than the layer above. This is a potential aquifer target for hand-wall drilling (Table 5).

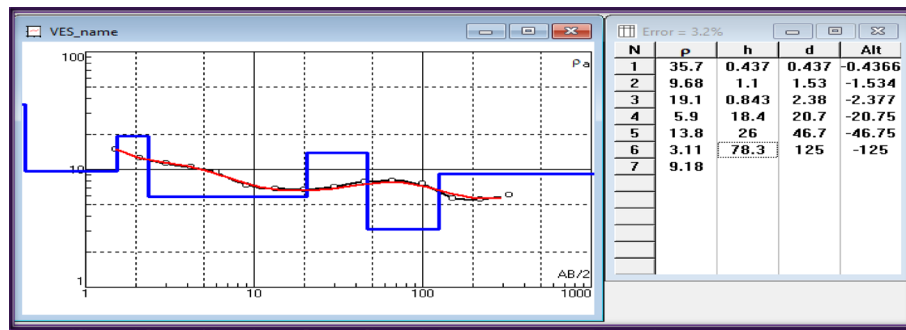


Fig. 9 Vertical electrical resistivity variation curve at cheffa

Table 5 Description of resistivity, thickness, depth and its interpretation at Cheffa

layer	Resistivity ($\Omega.m$)	Thickness (m)	Depth (m)	Geological interpretation
1	35.7	0.44	0.44	Topsoil
2	9.68	1.1	1.53	Clay
3	19.1	0.84	2.38	less clayey, more porous zone
4	5.9	18.4	20.7	Clay
5	13.8	26.0	46.7	sandy layer
6	3.11	78.3	125	Clay Layer
7	9.18			

Dawa Cheffa Kechure kebele, the Sand and loose dry soil comprise the subsurface up to 18.4 m. Below these layers, which may have altered to a sandy layer up to 46.7 m, these strata may be signs of groundwater accumulation.

4.2 Shallow ground water resource availability

The analysis of pumping discharge data from the Chefa borehole system reveals a high degree of consistency across the 25 sampled boreholes, with a predominant discharge rate of 16.6 L per second (l/s). The aggregate pumping capacity for the system were 415 L/s, equivalent to an annual yield of approximately 130,994 cubic meters (Table 6). This cumulative discharge emphasizes the significant and reliable irrigation potential afforded by the borehole infrastructure within the Cheffa district.

Based on hydrogeological data from the Habru (Girana) hand-dug well field, each of the three boreholes demonstrates a sustainable yield of 4 L/s. This corresponds to an estimated annual water volume of between 28.8 and 57.6 thousand cubic meters (m^3) per well. Collectively, the well field provides a combined discharge of 12 L/s, yielding an annual water supply of approximately 115,200 m^3 . The available data indicate a substantial groundwater potential within the Girana area. The yield is sufficient to support planned irrigation activities across a significant command area, confirming the viability of the resource for agricultural development but the farmers currently use for domestic needs and small-scale gardening.

The randomly sampling of 31 Hand-dug Well Data, all 31 wells show a pumping discharge of 16.6 L/s each, with the cumulative pumping discharge summing up to 514.6 L/s (108882.2 m^3/yr) (Table 8). This indicates a strong irrigation potential from the dug-well system in Kobo, providing a reliable and uniform discharge capacity across the wells.

Table 6 Annual consumption of irrigation water volume at Dawa Chefa hand-dug well Information

Bore-hole ID	X	Y	Z	Total drill depth (m)	dynamic water level (m)	Pumping discharge rate (l/s)	Farm size (ha)	Irrigation frequency	Well yield (m ³)/year
1	589,041	1,193,579	1438	20	4	16.6	7	9	12908.16
2	588,188	1,194,252	1430	53	4	16.6	5	9	7529.76
3	589,313	1,199,354	1495	50	12	16.6	1	7	1673.28
4	591,159	1,185,440	1420	25	6	16.6	2	10	4780.8
5	591,350	1,185,236	1420	22	7	16.6	1	9	2151.36
6	590,952	1,185,418	1421.9	20	6	16.6	2	9	4302.72
7	590,790	1,185,169	1421.1	20	5	16.6	3	9	6454.08
8	590,917	1,185,119	1420	20	6	16.6	2	10	4780.8
9	590,745	1,185,074	1421.7	22	7	16.6	2.5	11	6573.6
10	590,921	1,184,938	1420	22	8	16.6	3	10	7171.2
11	590,898	1,184,773	1421	20	6	16.6	2.5	9	5378.4
12	590,668	1,184,881	1423.5	20	6	16.6	2	9	4302.72
13	590,255	1,184,939	1418.7	20	6	16.6	3	9	6454.08
14	590,217	1,185,171	1421	20	6	16.6	3	9	6454.08
15	589,910	1,184,870	1422	20	6	16.6	2	10	4780.8
16	589,718	1,185,039	1420.8	20	6	16.6	2	9	4302.72
17	590,158	1,184,381	1419.1	20	5	16.6	2	10	4780.8
18	590,750	1,184,395	1418.6	25	7	16.6	2	10	4780.8
19	589,993	1,186,572	1424.9	22	6	16.6	2	10	4780.8
20	590,087	1,186,421	1423	25	5	16.6	1.5	10	3585.6
21	590,562	1,186,245	1424	22	6	16.6	2	10	4780.8
22	590,116	1,185,967	1423.1	20	5	16.6	1.5	9	3227.04
23	590,135	1,185,517	1421.7	25	6	16.6	2.5	9	5378.4
24	590,288	1,185,268	1421.8	20	5	16.6	2	9	4302.72
25	589,764	1,185,801	1423.2	22	6	16.6	2.5	9	5378.4
						415	61	234	130993.92

4.3 Irrigation potential land

The hydrological and geological characteristics of the Kobo Girana and Cheffa districts, shallow groundwater offers a significant potential for irrigation development. The areal coverage of kobo valley is 1439 km² [12]. Kobo Aboure Kebele contains 448 hectares of potential cropland suitable for shallow groundwater utilization, at present day 220 hectares are covered by shallow groundwater withdrawal, which now benefits 95 farmers. Mesena Kebele, Dawa Chefa Woreda, a potential total land 214 hectors of land suitable for agricultural development by the hand dug well operation, only 42 farmers now employ 150 hectares for profitable shallow groundwater utilization (Fig. 10). Girana Kebele possesses an estimated 200 hectares of land with agricultural potential that will comfort shallow groundwater abstraction, but because of widespread urbanization, the resource is not used for irrigation. During the dry season, the shallow groundwater is utilized for domestic needs and small-scale gardening.

5 Conclusion and recommendation

5.1 Conclusion

Shallow groundwater is used as a source of water for irrigation farming for the last ten years. However, Despite the development of shallow groundwater irrigation, opportunities remain to improve irrigation efficiency and expand irrigated areas. To effectively address this issue, farmers should adopt improved shallow groundwater extraction

Table 7 Annual consumption of irrigation water volume at habru and kobo worda hand-dug well Information

Borehole	X	Y	Z	Total drill depth (m)	dynamic water level (m)	Pumping dis-charge (l/s)	Farm size (ha)	irrigation frequency	Annual consumption (m3)
1	578,984	1,280,234	1451	8	3	4	0.5	12	28.8
2	578,684	1,280,095	1427	8	4	4	1	11	57.6
3	578,293	1,280,034	1434	8	4	4	0.5	12	28.8
4	575,343	1,334,815	1393	30	6	16.6	2.5	9	4302.72
5	575,482	1,334,854	1395	30	6	16.6	2.5	9	4302.72
6	575,545	1,335,142	1390	40	7	16.6	1.5	7	2509.92
7	575,443	1,335,255	1384	30	7	16.6	2	7	2928.24
8	575,529	1,335,368	1375	35	6	16.6	2	7	2928.24
9	575,609	1,335,361	1364	35	4	16.6	2	9	3764.88
10	576,075	1,335,314	1378	30	5	16.6	3	7	4183.2
11	576,270	1,335,248	1361	30	5	16.6	1	9	2151.36
12	576,439	1,335,342	1360	30	5	16.6	1	7	1673.28
11	576,319	1,335,456	1360	35	6	16.6	2	9	3764.88
12	576,018	1,336,035	1378	30	4	16.6	2	7	2928.24
13	575,635	1,335,189	1374	30	5	16.6	5	9	8366.4
14	575,616	1,335,037	1373	30	6	16.6	2	7	2928.24
15	575,560	1,334,846	1377	30	4	16.6	2	8	3346.56
16	575,793	1,334,953	1370	30	2	16.6	2	8	3346.56
17	574,019	1,335,136	1396	20	8.5	16.6	0.5	8	956.16
18	574,023	1,335,959	1385	24	3	16.6	2	9	3764.88
19	573,930	1,335,931	1385	30	3	16.6	2	8	3346.56
20	573,758	1,336,030	1395	30	5	16.6	3	7	4183.2
21	573,233	1,336,502	1399	30	7	16.6	2	9	3764.88
22	573,092	1,336,445	1398	30	8	16.6	2	9	3764.88
23	573,185	1,336,318	1395	30	8	16.6	2	8	3346.56
24	573,351	1,336,168	1395	30	9	16.6	2	9	3764.88
25	573,270	1,335,935	1402	35	9	16.6	2	8	3346.56
26	573,125	1,336,017	1390	35	9	16.6	2	8	3346.56
27	573,120	1,336,031	1405	32	9	16.6	3	8	4780.8
28	573,027	1,336,021	1406	45	10	16.6	1	6	1434.24
29	572,909	1,336,136	1397	42	9	16.6	2	9	3764.88
30	572,900	1,336,022	1406	42	9	16.6	2	9	3764.88
31	572,948	1,335,928	1399	42	10	16.6	3	8	4780.8
32	573,019	1,335,874	1403	42	10	16.6	2	8	3346.56
Total						526.6	67		108997.9

techniques. Such sustainable methods would improve the system's dependability and efficiency, allowing for profitable and successful year-round farming. Improved ground-water management may increase cropping intensity where groundwater availability is sufficient. Kobo Woreda, all 95 of the manually dug wells constructed over the preceding ten years are still in functioning. In contrast, 78 boreholes have been developed in the Dawa Cheffa Woreda over the last 7 years all are functional. Shallow groundwater development is mostly influenced by the topographic elevation gradient in the region. At the watershed outlet, groundwater is readily available at shallow depths. However, manual excavation becomes difficult as the watershed moves upslope since deeper drilling coincides with the increasing slope. Water extraction also requires higher horsepower pumps, which raises prices to levels that are often out of reach for local farmers. The primary advantages of shallow groundwater development on private agriculture are its

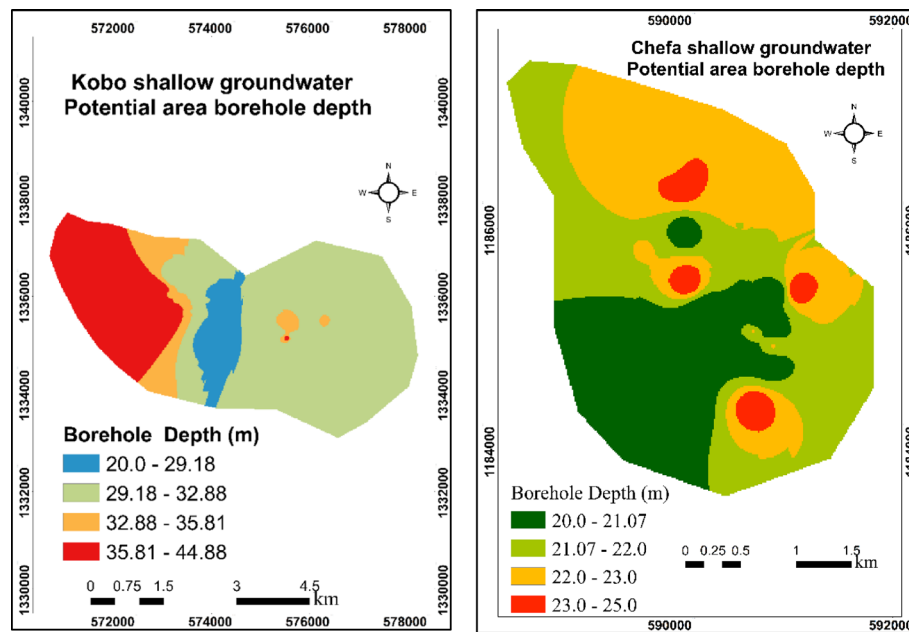


Fig. 10 Shallow groundwater potential map borehole depth

direct contribution to increased farm profitability, simplicity of operation, and potential for technological advancement.

5.2 Recommendation

Prioritization will be emphasis for groundwater development in Kobo and Girana, where thick, high-yield aquifers (coarse sand/gravel and weathered zones) occur at accessible depths (8–28 and 21–39 m, respectively), supporting sustainable irrigation abstraction. In Gerado, shallow aquifers (< 20 m) are perched and insufficient for irrigation, deeper layers are dominated by low-permeability clay or weathered basalt. In Jari, cobble/boulder deposits obstruct manual drilling, focus should instead be on exploring fractured basalt zones for potential deeper yields. Standardize borehole construction to target the most conductive saturated zones identified in VES profiles (e.g., layers 3–4 in Kobo, layers 5–6 in Girana). Maintain pumping rates at ≤ 16.6 L/s to avoid over-extraction and ensure long-term sustainability. Continuous monitoring wells and water-level loggers at key sites (Kobo, Girana, Cheffa) is needed to track dynamic water levels, salinity trends, and seasonal fluctuations, enabling adaptive management and early warning of depletion. Encourage farmer cooperatives for shared well operation and maintenance, especially in high-yield zones, to optimize water use efficiency and reduce individual drilling risks. Use resistivity-derived lithological maps to guide land-use planning, aligning crop selection and irrigation scheduling with aquifer recharge capacity and soil moisture retention characteristics. A targeted capacity-building program is needed to help farmers switch from traditional hand-dug methods to contemporary, light appropriate low-cost mechanized drilling technologies in order to advance sustainable irrigation practices.

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Author contributions

1. Gebeyaw Mollaw Yimer: Conceptualization, methodology, validation, investigation, writing - original draft. 2. Critical validation was performed by Gashaw Beza Hussien and Sisay Dessale Abate, they actively participated in reviewing and editing the manuscript. All authors read and approved the final manuscript.

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Data availability

The datasets utilized and analyzed in this study are accessible from the corresponding author upon reasonable request.

Declarations**Ethics approval and consent to participate**

This study did not involve human data or human tissue. All field work and shallow aquifer characterization were conducted in accordance with institutional, national, and international guidelines and legislation. Consent to participate is Not applicable.

Consent for publication

Not applicable.

Competing interests

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